

BRACKNELL BEAUFORT, United Kingdom

WMO: 037630

Lat: 51.383N

Lon: 0.783W

Elev: 243

StdP: 14.57

Time Zone: 0.00 (EUW)

Period: 94-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.7	(c) 26.9	(d) 19.7	(e) 14.9	(f) 30.3	(g) 22.4	(h) 17.0	(i) 31.1	(j) 17.7	(k) 44.2	(l) 16.0	(m) 45.0	(n) 2.0	(o) 50	(p) 0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 18.4	(c) 81.5	(d) 65.0	(e) 77.4	(f) 63.3	(g) 74.1	(h) 62.3	(i) 67.2	(j) 76.5	(k) 65.2	(l) 74.0	(m) 63.6	(n) 71.5	(o) 5.8	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	88.9	71.0	62.0	83.7	68.8	60.4	79.0	66.5	31.7	75.9	30.2	74.0	29.0	71.8	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.2	(b) 13.4	(c) 11.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 19.2	(b) 69.6	(c) 2.4	(e) 19.2	(f) 69.6	(g) 2.4	(h) 2.3	(i) 17.5	(j) 71.3	(k) 16.1	(l) 72.7	(m) 14.8	(n) 74.0	(o) 13.0	(p) 75.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	40.1	42.1	44.6	47.3	53.0	58.2	62.4	63.1	57.0	52.0	46.3	40.7	
	DBStd	9.59	6.24	5.78	4.97	5.79	5.38	4.53	4.28	5.10	4.02	5.43	5.73	7.04	
	HDD50	1336	308	225	178	117	30	1	0	0	2	41	137	297	
	HDD65	5320	771	642	633	531	371	209	104	99	241	403	561	755	
	CDD50	1560	2	2	10	36	124	247	384	407	212	103	26	7	
	CDD65	71	0	0	0	0	0	6	23	41	1	0	0	0	
	CDH74	753	0	0	0	2	22	88	234	392	15	0	0	0	
	CDH80	159	0	0	0	0	0	16	39	104	0	0	0	0	
Wind	WSAvg	4.8	5.6	6.0	5.5	5.1	4.8	4.7	3.9	3.7	4.1	4.6	4.5	5.1	
Precipitation	PrecAvg	27.9	3.0	2.1	1.7	2.1	1.9	2.0	1.9	2.3	2.2	2.9	3.1	2.7	
	PrecMax	37.4	7.2	5.0	4.4	5.6	4.1	5.0	4.8	4.5	4.6	6.1	6.3	5.1	
	PrecMin	20.6	0.4	0.2	0.3	0.1	0.2	0.5	0.5	0.4	0.4	1.1	1.2	0.5	
	PrecStd	4.5	1.5	1.3	0.9	1.3	1.0	1.2	0.9	1.3	1.1	1.3	1.5	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.3	55.8	61.3	69.6	76.7	82.7	84.7	88.2	76.5	69.7	60.0	56.5	
		MCWB	51.1	50.0	51.0	55.9	59.5	64.0	66.8	68.4	64.0	61.6	56.5	53.4	
	2%	DB	52.8	53.9	57.9	64.7	73.0	76.3	80.0	82.8	70.2	64.2	57.9	54.9	
		MCWB	49.6	49.5	49.4	52.7	60.1	61.4	64.4	65.4	60.6	57.7	55.6	52.2	
	5%	DB	51.0	52.1	55.4	60.6	68.4	71.9	76.4	78.5	67.5	62.0	56.3	53.5	
		MCWB	48.1	48.4	49.1	51.0	57.8	60.2	63.7	64.7	58.8	57.0	53.8	51.2	
	10%	DB	49.4	50.4	53.0	57.2	63.5	68.8	73.3	73.9	64.8	60.1	54.6	51.6	
		MCWB	46.9	47.0	48.1	49.6	55.0	58.4	62.2	62.9	57.7	56.4	51.9	49.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	52.1	53.9	57.1	64.2	66.0	69.9	70.1	64.6	62.8	57.5	54.5
			MCDB	54.2	54.9	58.3	63.5	71.8	79.2	81.0	83.6	72.2	68.4	58.6	55.5
2%		WB	50.1	50.4	51.9	54.2	61.2	63.2	67.0	67.7	62.4	60.0	56.0	53.0	
		MCDB	52.1	52.9	55.0	60.9	70.0	72.5	75.7	77.4	68.3	62.6	57.5	54.5	
5%		WB	48.6	49.0	50.6	52.5	58.2	61.5	64.9	65.8	60.9	58.3	54.4	51.3	
		MCDB	50.6	51.5	53.5	58.3	66.7	69.6	73.3	74.5	65.3	60.8	56.1	53.1	
10%		WB	47.1	47.4	49.0	50.7	55.9	59.8	63.0	64.0	59.4	56.9	52.3	49.5	
		MCDB	49.2	50.1	52.2	56.1	62.5	66.8	71.0	72.0	63.1	59.6	54.0	51.6	
Mean Daily Temperature Range		5% DB	MDBR	9.7	11.4	13.7	15.8	16.5	17.0	17.7	18.4	14.9	13.4	10.4	9.1
			MCDBR	10.2	11.2	17.6	20.8	26.0	24.7	24.7	25.9	19.1	14.3	8.9	8.9
	MCWBR		9.2	9.0	11.2	11.1	14.3	12.5	12.0	10.7	10.3	9.4	7.3	8.6	
	5% WB	MCDBR	10.4	11.0	13.2	17.8	23.4	20.8	20.4	21.8	16.2	13.0	8.3	8.5	
		MCWBR	9.9	9.8	9.8	11.0	13.9	11.5	11.0	10.3	9.9	9.9	7.8	8.5	
Clear-Sky Solar Irradiance	taub		0.329	0.345	0.381	0.403	0.406	0.405	0.400	0.391	0.374	0.358	0.332	0.318	
	taud		2.365	2.371	2.265	2.224	2.255	2.284	2.307	2.349	2.391	2.418	2.409	2.351	
	Ebn at Noon		215	246	256	264	269	270	269	266	257	237	213	198	
	Edh at Noon		20	26	35	40	41	40	38	35	30	24	19	17	
All-Sky Solar Radiation	RadAvg		260	470	818	1267	1534	1698	1609	1339	1007	583	310	204	
	RadStd		21	61	111	144	96	172	154	108	84	52	35	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page